Technical University of Denmark

Advanced machine learning (02460)

Exam, May 14, 2024

Exam duration: 2 hours.

Aid: Only written works of reference are permitted.

IMPORTANT: Only information written on the pages 2–12 will be evaluated.

Info about you: *You must write your student number on all pages at the bottom-left.*

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Part I
Generative models

Exercise A Variational autoencoder (VAE)

Consider a standard deep latent variable model with latent variable $z \in \mathbb{R}$ and observed variable $\mathbf{x} \in \mathbb{R}^2$. Let z follow a standard Gaussian distribution, i.e., $p(z) = \mathcal{N}(z|0,1)$, and the distribution of $\mathbf{x}$ be given by

$$p(\mathbf{x}|z) = \mathcal{N}\left(\mathbf{x} \,\middle|\, \rho\left(\begin{bmatrix}1\\-1\end{bmatrix} z + \begin{bmatrix}1\\-1\end{bmatrix}\right), \mathbf{I}_2\right), \tag{1}$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix and $\rho(\mathbf{x}) = \max(0, \mathbf{x})$ is the ReLU activation function. Consider also a variational posterior given by

$$q(z|\mathbf{x}) = \mathcal{N}\left(z \,\middle|\, \mu = \begin{bmatrix}1 & -1\end{bmatrix}\mathbf{x} - 1, \sigma^2 = \frac{1}{4}\right). \tag{2}$$

Recall that the ELBO for a VAE can be written as a reconstruction term and a KL term, i.e.,

$$\mathrm{ELBO}(\mathbf{x}) = \mathbb{E}_{z\sim q(z|\mathbf{x})}[\ln p(\mathbf{x} \mid \mathbf{z})] - \mathrm{KL}[q(z \mid \mathbf{x}) \,\|\, p(z)]. \tag{3}$$

Also, recall that the probability density function for the k-variate Gaussian distribution with mean vector $\boldsymbol{\mu} \in \mathbb{R}^k$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{k\times k}$ is given by

$$\mathbf{x} \mapsto \frac{1}{\sqrt{(2\pi)^k|\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right), \tag{4}$$

and that the KL divergence between two *univariate* Gaussian distributions has the closed form

$$\mathrm{KL}[\mathcal{N}(\mu_1,\sigma_1^2) \,\|\, \mathcal{N}(\mu_2,\sigma_2^2)] = \ln\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}. \tag{5}$$

Question A.1: Assuming a one-sample Monte Carlo estimate of the reconstruction term, where the sampled value is $z' = 0$, what is the ELBO evaluated at $\mathbf{x}' = \begin{bmatrix}1\\0\end{bmatrix}$?

A ☐ $\mathrm{ELBO}(\mathbf{x}') = -\frac{3}{2} - \ln 4\pi$

B ☐ $\mathrm{ELBO}(\mathbf{x}') = -\frac{3}{8} - \ln 4\pi$

C ☐ $\mathrm{ELBO}(\mathbf{x}') = -\ln 2\pi$

D ☐ $\mathrm{ELBO}(\mathbf{x}') = \frac{3}{8} - \ln \pi$

Give a brief argument for your answer.

Exercise B Normalizing flows

Consider a normalising flow with random variables $\mathbf{u}, \mathbf{x} \in \mathbb{R}^2$, where the base distribution is given by

$$p_{\mathrm{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u}|\mathbf{0}, \mathbf{I}_2), \tag{6}$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix, and the density of $\mathbf{x}$ is given by

$$p_{\mathrm{x}}(\mathbf{x}) = p_{\mathrm{u}}(\mathbf{u})|\det J_T(\mathbf{u})|^{-1} \quad \text{where} \quad \mathbf{u} = T^{-1}(\mathbf{x}). \tag{7}$$

Let the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined as the composition $T = T_2 \circ T_1$, where

$$T_1\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}1\\2\end{bmatrix}, \det J_{T_1}\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = 4 \quad \text{and} \quad T_1\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \det J_{T_1}\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = 2,$$

and

$$T_2\left(\begin{bmatrix}1\\2\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}, \det J_{T_2}\left(\begin{bmatrix}1\\2\end{bmatrix}\right) = 2, \quad \text{and} \quad T_2\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = \begin{bmatrix}1\\0\end{bmatrix}, \det J_{T_2}\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = 3.$$

Recall that the probability density function for the k-variate Gaussian distribution is given by eq. (4), that $|\det J_T(\mathbf{u})|^{-1} = |\det J_{T^{-1}}(\mathbf{x})|$, and that

$$(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1} \quad \text{and} \quad \det J_{T_2 \circ T_1}(\mathbf{u}) = \det J_{T_2}(T_1(\mathbf{u})) \cdot \det J_{T_1}(\mathbf{u}). \tag{8}$$

Question B.1: What is $p_{\mathrm{x}}\left(\begin{bmatrix}1\\0\end{bmatrix}\right)$?

A ☐ $p_{\mathrm{x}}\left(\begin{bmatrix}1 & 0\end{bmatrix}^\intercal\right) = \frac{\exp(-5)}{16\pi}$

B ☐ $p_{\mathrm{x}}\left(\begin{bmatrix}1 & 0\end{bmatrix}^\intercal\right) = \frac{\exp(-5)}{12\pi}$

C ☐ $p_{\mathrm{x}}\left(\begin{bmatrix}1 & 0\end{bmatrix}^\intercal\right) = \frac{\exp(-4)}{16\pi}$

D ☐ $p_{\mathrm{x}}\left(\begin{bmatrix}1 & 0\end{bmatrix}^\intercal\right) = \frac{\exp(-4)}{12\pi}$

Give a brief argument for your answer.

Exercise C Denoising Diffusion Probabilistic Models

Consider a DDPM model of two-dimensional real-valued data with $T = 5$ latent variables, i.e., where $\mathbf{x}_0 \in \mathbb{R}^2$ is that data variable and $\mathbf{x}_1, \ldots, \mathbf{x}_5 \in \mathbb{R}^2$ are the latent variables. Let the variance schedule be given by $\beta_1 = \frac{1}{6}$, $\beta_2 = \frac{1}{6}$, $\beta_3 = \frac{1}{3}$, $\beta_4 = \frac{1}{3}$ and $\beta_5 = \frac{1}{2}$.

Recall that the forward process in the DDPM is a Markov chain that gradually adds Gaussian noise to the data using, i.e.,

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}). \tag{9}$$

Recall also, that if we marginalise over $\mathbf{x}_{1:t-1}$, we obtain a Gaussian

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t|\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}) \quad \text{where} \quad \bar{\alpha}_t = \prod_{\tau=1}^{t}(1-\beta_\tau) \tag{10}$$

Recall that the probability density function for the k-variate Gaussian distribution is given by eq. (4).

Question C.1: What is $q\left(\mathbf{x}_2 \middle| \mathbf{x}_0 = \begin{bmatrix} -6 \\ 12 \end{bmatrix}\right)$?

A ☐ $q\left(\mathbf{x}_2 \middle| \mathbf{x}_0 = \begin{bmatrix} -6 & 12 \end{bmatrix}^\intercal\right) = \mathcal{N}\left(\mathbf{x}_2 \middle| \begin{bmatrix} -1 & 2 \end{bmatrix}^\intercal, \frac{1}{36}\mathbf{I}\right)$

B ☐ $q\left(\mathbf{x}_2 \middle| \mathbf{x}_0 = \begin{bmatrix} -6 & 12 \end{bmatrix}^\intercal\right) = \mathcal{N}\left(\mathbf{x}_2 \middle| \begin{bmatrix} -1 & 2 \end{bmatrix}^\intercal, \frac{35}{36}\mathbf{I}\right)$

C ☐ $q\left(\mathbf{x}_2 \middle| \mathbf{x}_0 = \begin{bmatrix} -6 & 12 \end{bmatrix}^\intercal\right) = \mathcal{N}\left(\mathbf{x}_2 \middle| \begin{bmatrix} -5 & 10 \end{bmatrix}^\intercal, \frac{25}{36}\mathbf{I}\right)$

D ☐ $q\left(\mathbf{x}_2 \middle| \mathbf{x}_0 = \begin{bmatrix} -6 & 12 \end{bmatrix}^\intercal\right) = \mathcal{N}\left(\mathbf{x}_2 \middle| \begin{bmatrix} -5 & 10 \end{bmatrix}^\intercal, \frac{11}{36}\mathbf{I}\right)$

Give a brief argument for your answer.

Part II

Representation learning and identifiability

Exercise D Pull-back metrics

Consider the generative model $f : \mathbb{R}^d \to \mathbb{R}^D$

$$f(\mathbf{x}) = \mathbf{W}_2 \sigma\left(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2, \tag{11}$$

where $\mathbf{W}_1 \in \mathbb{R}^{D \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{D \times D}$ are full-rank matrices ($D \geq d$), and $\mathbf{b}_1 \in \mathbb{R}^D$ and $\mathbf{b}_2 \in \mathbb{R}^D$ are bias vectors. Here σ denotes any smooth (elementwise) activation function.

Let $\dot{\sigma}(\cdot)$ denote the Jacobian of the activation function. Then consider the following four matrices:

$$\begin{aligned}
\mathbf{G}_{\mathbf{x}}^{(A)} &= \mathbf{W}_1^\intercal \dot{\sigma}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)^\intercal \mathbf{W}_2^\intercal \mathbf{W}_2 \dot{\sigma}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \mathbf{W}_1, \\
\mathbf{G}_{\mathbf{x}}^{(B)} &= \mathbf{W}_2^\intercal \dot{\sigma}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)^\intercal \mathbf{W}_1^\intercal \mathbf{W}_1 \dot{\sigma}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \mathbf{W}_2, \\
\mathbf{G}_{\mathbf{x}}^{(C)} &= \mathbf{W}_1^\intercal \dot{\sigma}(\mathbf{W}_1 \mathbf{x})^\intercal \mathbf{W}_2^\intercal \mathbf{W}_2 \dot{\sigma}(\mathbf{W}_1 \mathbf{x}) \mathbf{W}_1, \\
\mathbf{G}_{\mathbf{x}}^{(D)} &= \mathbf{W}_2^\intercal \dot{\sigma}(\mathbf{W}_1 \mathbf{x})^\intercal \mathbf{W}_1^\intercal \mathbf{W}_1 \dot{\sigma}(\mathbf{W}_1 \mathbf{x}) \mathbf{W}_2.
\end{aligned}$$

Question D.1: The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{J}_{\mathbf{x}}^\intercal \mathbf{J}_{\mathbf{x}}$, where $\mathbf{J}_{\mathbf{x}}$ is the Jacobian of f. Which one of the following statements is true?

A ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(A)}$.

B ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(B)}$.

C ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(C)}$.

D ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(D)}$.

Give a brief argument for your answer.

Exercise E Curve energy

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \begin{pmatrix} x_1^2 \\ x_1^2 + 2x_2 \\ x_1^2 + x_2^2 \end{pmatrix}. \tag{12}$$

Furthermore, consider the two latent curves

$$c_1(t) = t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad c_2(t) = t \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{13}$$

Question E.1: Derive the pull-back metric $\mathbf{G}_\mathbf{x} = \mathbf{J}_\mathbf{x}^\intercal \mathbf{J}_\mathbf{x}$, where $\mathbf{J}_\mathbf{x}$ denotes the Jacobian of f at $\mathbf{x}$.

Question E.2: Let the energy of a curve be given by

$$\mathcal{E}(c) = \int_0^1 \dot{c}(t)^\intercal \mathbf{G}_{c(t)} \dot{c}(t) \mathrm{d}t, \tag{14}$$

where $\dot{c}(t) = \frac{\mathrm{d}c(t)}{\mathrm{d}t}$ is the curve derivative. Evaluate the energy of the curves c_1 and c_2.

$\mathcal{E}(c_1) =$ _____ $\mathcal{E}(c_2) =$ _____

Sketch your derivations below.

Exercise F Integration

Consider the metric over $\mathbf{x} = \left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) \in \mathbb{R} \times \mathbb{R}_+$

$$\mathbf{G}_{\mathbf{x}} = \begin{pmatrix} 1/x_2 & 0 \\ 0 & 1/x_2 \end{pmatrix}. \tag{15}$$

Let Ω denote the region $[-1, 1] \times (0, 1]$.

Question F.1: Derive the volume measure $\mathrm{d}\mathcal{M}(\mathbf{x}) = \sqrt{\det \mathbf{G}_{\mathbf{x}}}\mathrm{d}\mathbf{x}$.

$\mathrm{d}\mathcal{M}(\mathbf{x}) =$ _____

Sketch your derivations below.

Question F.2: Evaluate the integrals

$$I_1 = \int_\Omega x_1^2 \, x_2 \mathrm{d}\mathcal{M}(\mathbf{x}), \tag{16}$$

$$I_2 = \int_\Omega x_1^2 \, x_2^2 \mathrm{d}\mathcal{M}(\mathbf{x}). \tag{17}$$

$I_1 =$ _____ $I_2 =$ _____

Sketch your derivations below.

Exercise G Information geometry

In information geometry, the *Bregman divergence* for comparing two distributions p and q is given as

$$D_F(p,q) = F(p) - F(q) - \langle \nabla F(q), p - q \rangle, \tag{18}$$

for some function $F : \mathcal{H} \to \mathbb{R}$, where $\mathcal{H}$ denotes the space of distributions. Here $\nabla F(q) = \begin{pmatrix} \partial F / \partial q_1 \\ \vdots \\ \partial F / \partial q_D \end{pmatrix}$.

Consider the two probability vectors

$$p = \begin{pmatrix} p_1 \\ \vdots \\ p_D \end{pmatrix} \qquad \text{and} \qquad q = \begin{pmatrix} q_1 \\ \vdots \\ q_D \end{pmatrix}, \tag{19}$$

where $p_i > 0$, $q_i > 0$, and $\sum_i p_i = \sum_i q_i = 1$.

Let F evaluate the negative entropy,

$$F(p) = \sum_{i=1}^{D} p_i \log(p_i). \tag{20}$$

Question G.1: Show that the Bregman divergence D_F is then $\mathrm{KL}(p \| q) = \sum_{i=1}^{D} p_i \log\left(\frac{p_i}{q_i}\right)$.

Part III
Graph models

Exercise H Shallow embeddings

In this exercise we will use the squared distance decoder $d_{u,v} = \mathrm{DEC}(\boldsymbol{z}_u, \boldsymbol{z}_v) = \|\boldsymbol{z}_u - \boldsymbol{z}_v\|_2^2$. We will use a 2-dimensional embedding space, and refer to the coordinates of the embedding as $\boldsymbol{z}_v = [x, y]^\top$.

Question H.1: Let us assume that node u is embedded as $\boldsymbol{z}_u = [1, 0]^\top$. Where could node v be embedded, if the decoder of the node pair evaluates to $d_{u,v} = 0.25$?

A ☐ On a circle of radius 0.5 around $\boldsymbol{z}_u$, i.e. $(x-1)^2 + y^2 = 0.5^2$

B ☐ On a circle of radius 1 around the origin, i.e. $x^2 + y^2 = 1$

C ☐ On the line $x = -1$

D ☐ On the line $y = 0$

E ☐ None of the above

Show the main steps of your derivation.

H

Exercise I Graph Laplacians

The random walk graph Laplacian is defined as

$$\boldsymbol{L}_{\mathrm{rw}} = \boldsymbol{I} - \boldsymbol{D}^{-1}\boldsymbol{A}$$

where $\boldsymbol{A}$ is the adjacency matrix and $\boldsymbol{D}$ is the degree matrix. Consider a graph with the following random walk Laplacian:

$$\boldsymbol{L}_{\mathrm{rw}} = \begin{bmatrix} 1 & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

where we will label the nodes a–e in the order of the rows of the Laplacian.

Question I.1: Sketch the graph

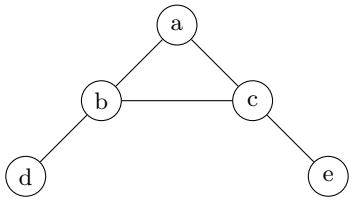
Question I.2: We now consider a random walk on the graph, starting at node a. What is the probability p that we end up in node b after exactly 3 steps of the random walk?

$p =$ _____

Give a brief argument for your answer.

Exercise J Weisfieler-Lehman test

Consider the following graph



Question J.1: We consider running the Weisfeiler-Lehman algorithm:

- Assign initial labels (node degree)
$$l_v^{(0)} = d_v$$
- Iteratively assign new labels to each node by hashing the multi-set of the neighbors' labels
$$l_v^{(i)} = \textsc{hash}(\{\{l_u^{(i-1)} \forall u \in \mathcal{N}(v)\}\})$$

Which of the following statements are true. **Select all that apply.**

A ☐ $l_{\mathrm{a}}^{(0)} = 3$.

B ☐ $l_{\mathrm{b}}^{(0)} = 3$.

C ☐ $l_{\mathrm{a}}^{(1)} = \textsc{hash}(\{\{2, 3\}\})$.

D ☐ $l_{\mathrm{b}}^{(1)} = \textsc{hash}(\{\{1, 2, 3\}\})$.

E ☐ $l_{\mathrm{d}}^{(1)} = \textsc{hash}(\{\{2, 3, 3\}\})$

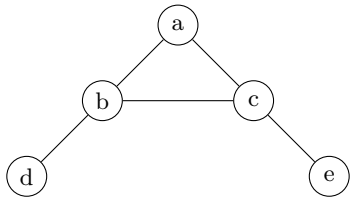
Question J.2: The Weisfeiler-Lehman algorithm halts when the assigned labels no longer generate a different partitioning of the nodes. After how many rounds does the Weisfeiler-Lehman algorithm halt for the given graph?

Number of rounds = _____

Give a brief argument for your answer.

Exercise K Graph convolution

Consider the following graph



The following code prints the adjacency matrix and computes its eigendecomposition (the nodes are ordered *a–f*.)

```
>>> print(A)
[[0. 1. 1. 0. 0.]
 [1. 0. 1. 1. 0.]
 [1. 1. 0. 0. 1.]
 [0. 1. 0. 0. 0.]
 [0. 0. 1. 0. 0.]]

>>> l, U = np.linalg.eigh(A)

>>> print(np.round(l, 2))
[-1.62 -1.3  -0.    0.62  2.3 ]

>>> print(np.round(U, 2))
[[-0.    0.65 -0.58  0.    0.49]
 [-0.6  -0.43  0.    0.37  0.57]
 [ 0.6  -0.43  0.   -0.37  0.57]
 [ 0.37  0.33  0.58  0.6   0.25]
 [-0.37  0.33  0.58 -0.6   0.25]]
```

A graph convolution with a finite-length filter can be written in the vertex domain, and in the spectral domain as:

$$\boldsymbol{y} = \sum_{k=0}^{N} h[k]\boldsymbol{A}^k\boldsymbol{x} = \boldsymbol{U}\underbrace{\left(\sum_{k=0}^{N} h[k]\boldsymbol{\Lambda}^k\right)}_{\boldsymbol{M}}\boldsymbol{U}^*\boldsymbol{x}$$

where we have defined the Fourier-domain filter matrix

$$\boldsymbol{M} = \sum_{k=0}^{N} h[k]\boldsymbol{\Lambda}^k$$

Question K.1: Consider a filter defined as

$$h[k] = \begin{cases} 1 & k = 0 \\ -1 & k = 1 \\ 0 & \text{otherwise,} \end{cases}$$

Determine the first entry on the diagonal of the Fourier-domain filter matrix $\boldsymbol{M}$. Fill in the entry below, ignore the remaining entries.

$$\boldsymbol{M} = \begin{bmatrix} \underline{\qquad} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

K